



Report for the Core Project

Microscopic model of avalanches and oscillations
in neural networks

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1 Introduction

Interactions of many individual units can lead to intriguing collective phenomena, be it superconductivity, flocking behavior of birds, volatile behavior of stock markets, or the complex activity of neuronal networks. Networks of neurons found in living tissues exhibit various patterns of self-sustained continuous activity such as oscillations, propagating waves, and *neuronal avalanches* - which are spatio-temporal cascades of neural activity that exhibit power-law statistical distribution of their size and duration. Intuitively, avalanches correspond to cascade-like dynamics present in earthquakes, mountain avalanches, nuclear reactors and neuronal networks: a “unit” of the system receives energy from its surrounding, and if above a certain “firing” threshold, distributes it back to the system, causing a propagating avalanche of “firing” in its neighborhood. Such behavior does not show a characteristic size or duration but is present at all system scales in the same manner, and since the power-law function is the only mathematical function with no characteristic scale, this leads to a power-law statistical distribution of avalanche sizes and durations.

Recent studies suggest that neuronal avalanches correspond to dynamics of a system at the *critical state* [1], which has been previously suggested theoretically [2] since systems at criticality exhibit properties that could be advantageous to cortical function, such as:

- long-range correlations, important for achieving a coordinated collective behavior across neurons,
- large variability in achievable spatio-temporal patterns, allowing a large dynamical range, thus a possibly rich(er) representation of information,
- maximization of information transmissions through the network under the constraint of marginal stability against runaway excitations (as, for example, in epilepsy) [1].

Although current observations of neuronal avalanches and long-range correlations [1,3] across spatial scales and systems seem to support the hypothesis of criticality, their existence and role in neuronal networks is still not fully understood, and many questions remain. For example, how does one explain the *simultaneous* existence of the scale-specific neural oscillations and the scale-invariant neural avalanches in the brain? To investigate such questions, we have developed a computational model of a neural network based on the Ising model and have investigated basic microscopic features leading to the emergence of both oscillations and scale-free avalanches observed in our model.

2 Model

Based on the previous work [4], the model we use is a classical Ising model with the addition of negative feedback between the external field and the magnetization. The Ising model – historically introduced as a microscopic model of magnetism and now a favorite toy model of statistical physicists – consists of a collection of magnetic spins arranged in a square lattice, placed in an external

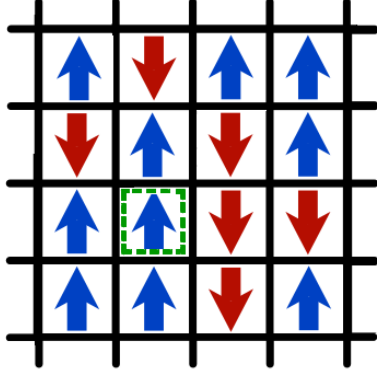


Figure 1: A square lattice with 16 spins, with 10 spins up and 6 spins down, giving a total magnetization of $m = 10 - 6 = 4$.

For $B = 0$, $J = 1$ and $kT = 2$, the spin marked with the green square has an energy of: $E = -(1) * (1) * (1 + 1 + 1 - 1) = -2$, and energy of 2 if it were flipped. As the change in energy would be +4, the probability of such flip would be: $\exp(-4/2) = \exp(-2) = 0.14$. Image adapted from Wikimedia Commons user *kismelec*, distributed under license CC BY-SA 3.0.

magnetic field and held at a certain temperature (Fig. 1). Spins can be aligned either up or down – equivalent to a neuron being active or not – and they flip their orientation depending on the interplay between random thermal fluctuations, which tend to create disorder, and magnetic fields of their neighboring spins and external magnetic field, which tend to create order in the orientation of spins. The (relative) probability of flipping from some spin configuration S_1 to configuration S_2 is given by the Boltzmann's equation:

$$p(S_1 \rightarrow S_2) = e^{-\Delta E_{12}/k_B T} \quad (1)$$

where $\Delta E_{12} = E(S_2) - E(S_1)$ is the change in system's energy due to the spin flip, T is the absolute temperature of the system, and k_B is the Boltzmann constant.

The overall spin orientation in the system can be expressed using the total magnetization, which is a sum of individual spin values: $m = 1/N * \sum \sigma_{ij}$, where $\sigma_{ij} = 1$ (spin up or down), N is the total number of spins, and where indices refer to some position $\sigma(i, j)$ in the lattice. Similarly, the total energy of the system is a sum of energies of individual spins: $E = \sum E_{ij}$, where the energy of a single spin σ_{ij} is:

$$E_{ij} = -B \sigma_{ij} - \sum_{mn=\langle ij \rangle} J \sigma_{ij} \sigma_{mn} \quad (2)$$

where B is strength of the external magnetic field, J is strength of the magnetic field created by a neighboring spin (also called the spin-spin coupling strength), and where the summation is performed over all direct neighbors of spin σ_{ij} (up, down, left, right). Qualitatively, the expression above results in a system that has lower energy, and thus more probability to be observed, if the spins are aligned to the external field and if the neighboring spins are mutually (oppositely) aligned when $J > 0$ ($J < 0$).

The overall state of the system, however, also depends on the temperature: at low temperatures, the tendency to align neighboring spins will dominate, leading to an ordered, strongly magnetized (ferromagnetic) state, while at high temperatures, the thermal fluctuations will dominate, leading to an unordered, weakly magnetized (paramagnetic) state (assuming $B = 0$). The (phase) transition between these two states occurs at a specific value of the temperature $T = T_c$, the critical temperature, and such a system is said to be in a critical state. In this state, the system exhibits peculiar behaviors mentioned earlier: long-range correlations and power-law scaling emerge in various thermodynamic quantities and spins clusters stop showing a characteristic size i.e. they start following power-law statistics.

Here we have studied the dynamics of a classical Ising model, but with the addition of negative feedback between the total magnetization and the external magnetic field. Computationally, the random nature of spin flips dictated by the Boltzmann's equation is implemented using the Metropolis algorithm:

- At each iteration step:
 - choose a random lattice position (i,j)
 - calculate the change in system's energy $\Delta E(i,j)$ if
 - spin σ_{ij} were to be flipped
 - if $\Delta E \leq 0$, “accept” the flip
 - if $\Delta E > 0$, “accept” the flip with a probability $e^{-\Delta E/(k_B T)}$, otherwise “reject” it
 - update and save variables (m, B)

where the external magnetic field, which influences the value of ΔE , is initially set to zero and is updated at each step according to: $B^{it+1} = B^{it} - m^{it}/N\tau$, where τ is the timescale of the negative feedback, N is the number of spins in the system, and all variables are expressed in arbitrary units.

3 Results and further work

With a negative feedback applied, a simple linear dynamical system, such as a pendulum or a point mass, would reach a stable equilibrium state and would be kept close to it (biologically, this can be interpreted as a homeostatic response). The Ising model, however, is a complex dynamical system due to its various interactions and phase transition, which leads to noticeably different behavior.

At low temperatures (ferromagnetic state), we have observed self-sustained oscillations (Fig. 2, left) which can be explained as follows: when spins are aligned up, the external field will begin to align downwards, which will cause some spins to flip their orientation, and after a few iterations, the combined effect of even stronger downwards field and neighboring spins being flipped will

cause a rapid succession of flips in the whole system until all spins are oriented downwards, which is when the cycle repeats again. At high temperatures, we have obtained an expected noise-like behavior due to thermal fluctuations causing spin flips in the system, while exactly at the critical temperature, we obtained a behavior (Fig. 2, right) which we suspected to correspond to avalanche dynamics.

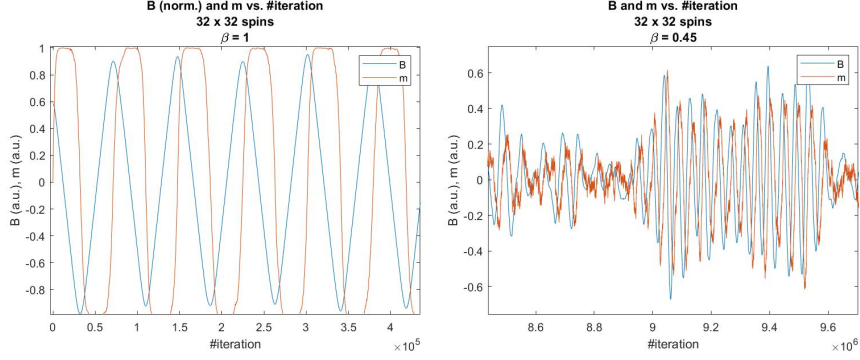


Figure 2: Plot of magnetization (m) and external magnetic field (B) versus number of iterations (“time”) obtained for a 32×32 spin system below the critical temperature (left) and at the critical temperature (right). Below the critical temperature, the system shows an oscillatory behavior with a well-defined period, while at the critical temperature, the system seems aperiodic in nature and the presence of random noise (thermal fluctuations) becomes more evident, which suggests that avalanche dynamics may be present.

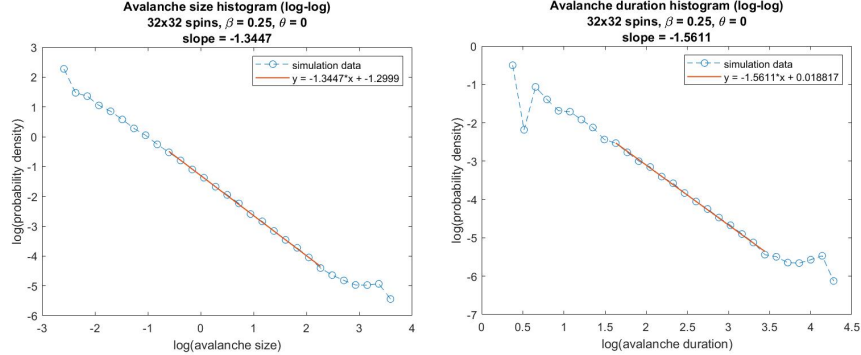


Figure 3: Log-log plots of avalanche size/duration probability density vs. size/duration for a 32×32 fully connected spin system at the critical temperature $\beta_c = (k_B T_c)^{-1} = 0.25$. For both the avalanche size (left) and duration (right), the linear region corresponds to a power-law scaling corresponding to the avalanche dynamics, while the small peak at large sizes / long durations corresponds to the presence of oscillatory behavior, therefore hinting at simultaneous existence of neuronal avalanches and oscillations in our system. The fitted values of power-law exponents are 1.35 for avalanche sizes and 1.56 for avalanche durations, and the fitting was performed over the linear region of the plot (red line). In a fully connected model all spins are considered as neighboring when calculating the energy, i.e. neighbors of a spin include not only the direct, but all other spins in the system.

To test whether scale-free avalanches are present in our system, we analyzed the histograms of avalanche sizes and durations, where we defined the avalanche duration as the time between two zero-crossings and the avalanche size as the area under the magnetization curve between two zero-crossings. A log-log plot of such histograms (Fig. 3) shows a linear region, that is, a power-law scaling which is characteristic for avalanche dynamics, and a small peak at large sizes / long durations which corresponds to the oscillatory dynamics, therefore showing a simultaneous existence of neural avalanches and oscillations in our model.

In the next step, we have compared our numerical results with the statistics of MEG (magnetoencephalography) measurements obtained from patients and the preliminary results show that power-law exponents (slopes) from in vivo data are higher than the ones obtained in the simulations. This mismatch can be attributed in part to the fact we analyzed the signal itself and not its envelope, which is a more common and correct data analysis method (see e.g. [3]), and to the fact that our model parameters were freely chosen and not fitted to the data in any manner.

To explore the influence of model parameters on the histograms and fitted power-law exponents, we have changed various parameters such as the temperature (Fig. 4), system size, avalanche detection threshold, and the feedback timescale τ . Current results indicate that the feedback timescale has the strongest influence among these model parameters, though further work will be needed to properly identify crucial parameters influencing the values of power-law exponents.

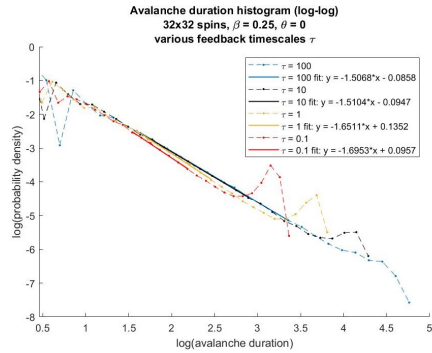


Figure 4: Log-log plot of avalanche duration histogram at various feedback timescales τ for a 32x32 fully connected spin system, where the black dotted line corresponds to the values used in previous figures. Values of fitted power-law exponents show a moderate variation, which could help us match the experimental and numerical results. Changes in the position and height of the rightmost peak suggest that oscillatory behavior become more prominent and faster as the feedback timescale is decreased, that is, as the feedback becomes faster and stronger.

In further work, we will explore the effect of changing feedback's spatial scale from a global to a more local scale, since currently all spins are subject to the same global feedback, which is somewhat biologically unrealistic for the number of "neurons" (ca. 1000) considered in our model. Similarly, the square lattice used so far will be replaced by a more biologically realistic lattice/network since the network connectivity is expected to influence the avalanche statistics. With these changes, we aim to arrive at a simple yet powerful model capable of producing both the neural oscillations and avalanches and their corresponding statistics, and advance the research on the role of criticality in neural networks.

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